

Contents

1	On the correct choice of models	2
1.1	Simplicial complexes	2
2	Goldblatt-Thomason Theorem	5

1 On the correct choice of models

Polyhedra enjoy many nice visualization properties due to their 'small' nature: The picture most people -among those, the author- have in mind when talking about Polyhedra is a compact, connected subset of $\mathbb{R}^n$.

While this picture is amazing for reasoning and for applications, it gets in the way of the goal of this paper:

Recall

Theorem 1.1 (Goldblatt-Thomason).

A class of (Kripke) frames is modally definable if and only if:

- *It's closed under bounded morphic images,*
- *It's closed under (arbitrary) disjoint unions,*
- *It's closed under - and reflects ultrafilter extensions.*

Images do nothing to subtract from our understanding of Polyhedra, but that is the only non-issue:

A class closed under arbitrary disjoint unions must contain a non-compact non-connected space: Say we tried to generate a class from the polyhedron consisting of a single vertex, we can obtain an infinite discrete set of $\aleph_0$ vertices by taking the disjoint union of the point with itself $\aleph_0$ times.

An even bigger problem is the closure under ultrafilter extensions: Say we have a polyhedron with infinite cardinality $\aleph_0$ -which we must, as was argued just above-, its ultrafilter extension must have cardinality $2^{2^{\aleph_0}}$, which is too big to be a subset of $\mathbb{R}^n$ for any finite n .

Annoyingly enough, we expect those extension to be well-behaved dimension-wise (the ultrafilter extension an n -dimensional polyhedron should be n -dimensional too as they express the same logic) meaning that embedding it in $\mathbb{R}^\kappa$ for an appropriate cardinal κ wouldn't do us any favors, as we should always be able to find a finite-dimensional space (thus of smaller cardinality than our set) in which it's embeddable.

So, what do we propose as models?

1.1 Simplicial complexes

Simplicial complexes are the most widely used model of polyhedra: They give all the topological information about them with minimal size.

Definition 1.2 (Simplicial Complex).

*A **Simplicial complex** is a pair (V, Σ) such that $\Sigma \subseteq \mathcal{P}(V)$ is closed under taking subsets.*

*We call the elements of Σ **Simplices** and we say that τ is a **face** of σ -denoted $\tau < \sigma$ - whenever $\tau \subseteq \sigma$.*

We will often avoid talking about V and regard Σ as the fundamental object, since V can be extracted as its singletons.

*A **Map** of simplicial complexes $(V, \Sigma) \rightarrow (V', \Sigma')$ is a function $f : V \rightarrow V'$ such that the image of every simplex of Σ is a simplex of Σ' .*

This is a pretty standard combinatorial description, that has its geometric counterpart:

Definition 1.3 (Geometric realization).

Consider a simplicial complex Σ regarded as a preorder with its face relation. Construct a functor $G : \Sigma \rightarrow \mathbf{Top}$ as follows:

- $G(\sigma) = \Delta^{|\sigma|}$ (the simplex of order "cardinality of σ ")
- $G(\tau \leq \sigma) : \Delta^m \rightarrow \Delta^n$ is the embedding of one of its faces.

In this, we modify the existing structure a little: We call $\tilde{G}$ the functor $\tilde{G}(X) = I(G(X))$ where I denotes the **Relative interior**, that is, the interior of the face seen as immersed in the smallest possible real affine space containing it. as it's usually the case.

The **Geometric realization** of Σ is the colimit $|\Sigma| := \varinjlim \tilde{G}$.

In other words it's the quotient space $\frac{\coprod_{\sigma \in \Sigma} (G(\sigma))^o}{\sim}$ where $\sim$ identifies each point with its face embedding.

Here's a picture for clarity:

The abstract setting allows us not to bother about embedding the polyhedron in a bigger space.

We encounter an issue here: Those who are familiar with the literature on polyhedral semantics know that we want to consider polyhedrally shaped subsets contained in a face, but this abstract view a priori does not give us access to points of a face.

The abstract setting allows us to use an important theorem:

Theorem 1.4. *For any x in the geometric realization of a simplicial complex, there exists a unique simplex σ such that $x \in \tilde{G}(\sigma)$*

Proof. □

Let us define:

Definition 1.5 (Polyhedra).

Let Σ, Σ' be two simplicial complexes. We say that $\Sigma \sim_p \Sigma'$ (read: Σ and Σ' are **Polyhedrally Equivalent**) whenever they have the same realization¹

A **Polyhedron** P is an equivalence class of simplicial complexes through $\sim_p$.

We have a way to pass from simplicial complexes to topological spaces, now we need a way to go back.

We need a couple of checks:

Proposition 1.6 (Equivalence and triangulation). *Let Σ be a simplicial complex and $x \in |\Sigma|$ any point on its realization.*

Then there exists a complex Σ' such that $\Sigma' \sim_p \Sigma$ and x is the realization of a vertex of Σ'

Proof.

Consider the simplicial complex P made of the vertices of $|\Sigma|$ (in **Top**). This realizes to itself, thus being equivalent to $|\Sigma|$.

Now consider the unique face $\sigma = (y_1, \dots, y_n)$ whose relative interior contains x . Add the lines $(xy_i)_{i=1, \dots, n}$, then the triangles $(xy_i y_j)_{i < j=1, \dots, n}$, etcetera until the n -simplices $(xy_{i_1} \dots y_{i_{n-1}})_{i_1 < \dots < i_{n-1}=1, \dots, n}$

As everything is contained in the same face σ we obtain the same realization as P , thus the same realization as Σ . □

¹Here 'the same' is a bit of a vague concept: Asking for **equality** is -as usual- too restrictive but **homeomorphism** is a bit too loose (recall that a square and a triangle are very much homeomorphic. We will discuss what the appropriate notion of "sameness" is in a later section)

Proposition 1.7 ($\sim_p$ induces morphisms).

Let Σ and Σ' be two polyhedrally equivalent simplicial complexes, then there exists a simplicial complex Σ'' in the same equivalence class and a pair of monomorphisms

$$\Sigma \hookrightarrow \Sigma'' \hookleftarrow \Sigma'$$

Proof.

This is non other than the triangulation lemma: We simply repeat the process described in the lemma above for all points not in common between the two.

The original faces are maintained thus the two maps are simply inclusions. $\square$

Proposition 1.8 ($\sim_p$ extends on morphisms).

Let $f : \Sigma \rightarrow \Sigma'$ be a morphism of simplicial complexes. Then for all morphisms $P \in [\Sigma]_{\sim_p} \rightarrow Q \in [\Sigma']_{\sim_p}$ there exist $P' \in [\Sigma]_{\sim_p} \rightarrow Q' \in [\Sigma']_{\sim_p}$ extending the two morphisms as pictured in the diagram:

$$\begin{array}{ccc} \Sigma & \longrightarrow & \Sigma' \\ \downarrow & & \downarrow \\ P' & \dashrightarrow & Q' \\ \uparrow & & \uparrow \\ P & \longrightarrow & Q \end{array}$$

Proof.

We apply the previous lemma to construct P' and Q' .

Every vertex of P' either comes from Σ or from P , thus the map is well defined on vertices.

Now, if there exists a simplex in P' that doesn't map onto a simplex of Q' yadda yadda i'm sleepy it should be doable tho. $\square$

2 Goldblatt-Thomason Theorem